Nikhil Varma

Propulsion and GNC Engineer

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Sample resume. The person, companies and projects are fictional.

I design rocket engines and the guidance software that brings boosters home. Five years of hot fires, hop tests and flight code at a small launch company in Bengaluru.

EXPERIENCE

Senior Propulsion and GNC Engineer

Jul 2023 to Present

Halyard Launch Systems, Bengaluru

- Lead landing guidance and thrust vector control for the H2 reusable first stage, with a team of four engineers.
- Wrote the powered descent guidance that flew on the Heron hop vehicle: 11 flights, every one landed within 0.6 m of the target.
- Own the 6-DOF flight simulator and the hardware in the loop rig that every flight software release has to pass.
- Serve as test conductor for hot fires and hop tests, including the go and no-go calls on the count.

Propulsion Engineer

Aug 2021 to Jun 2023

Halyard Launch Systems, Bengaluru

- Designed the coaxial swirl injector and the regenerative cooling channels for Ember, the company's 35 kN methalox engine.
- Took Ember from first ignition to a 120 second full duration burn across 41 hot fires.
- Built the test stand data system: 64 channels at 10 kHz with a 5 ms hardware redline abort.

GNC Engineer

Jun 2020 to Jul 2021

Orbitwise Dynamics, Bengaluru

- Wrote the attitude determination and control software for Kite-1, a 3U Earth observation cubesat.
- Kite-1 detumbled in 2.5 orbits after deployment and held 0.4 degree pointing on imaging passes.

FLAGSHIP: HERON HOP VEHICLE

Halyard's 4.2 m test vehicle for proving reusable landing. Led its guidance, navigation and control: the convex descent planner, the navigation filter fusing IMU, GNSS and a laser altimeter, and the TVC loop steering a single throttling Ember engine. 11 flights, 11 landings; mean touchdown error 0.38 m; guidance re-plans at 10 Hz, each solve under 18 ms.

SELECTED PROJECTS

Ember 35 kN methalox engine	Regen cooled, 96 channels, 42 bar, throttles 100 to 40 percent; 120 s full duration burn.
Coaxial swirl injector	61 elements, water cold flows and hot fire tuning; 97.8% c* efficiency, no instability.
Powered descent guidance	Lossless convexification SOCP in C++ with no dynamic memory; 18 ms worst case solve.
Electromechanical TVC	Two axis ball screw gimbal, 6 degree range, 22 deg/s slew, 9 Hz bandwidth, 9 kg lighter.
Landing navigation filter	Error state Kalman filter (IMU, GNSS, laser altimeter); under 4 cm touchdown altitude error.
Boostback and entry burn planner	Shooting method planner with grid fin CFD tables; 7% less propellant than baseline.
Hot fire test stand DAQ	64 channels at 10 kHz, FPGA redlines, 5 ms redline to abort.
Cryogenic loading autosequencer	Interlocked state machine for LOX and methane; loading 38% shorter.
Kite-1 cubesat ADCS	B-dot detumble, MEKF, reaction wheel pointing to 0.4 degrees.
Sarus-3 hybrid sounding rocket	Student team of 22, captain; nitrous and paraffin hybrid to 3.1 km apogee.

PAPERS AND TALKS

Real time powered descent guidance on a radiation tolerant flight computer. Launch Vehicle GNC Conference 2025, Hyderabad. Best Paper, guidance track.

Correlating wall heat flux in additively manufactured regenerative cooling channels. Journal of Space Propulsion, Volume 14, 2023.

Eleven hops: what a small test vehicle taught us about landing. Invited talk, Bengaluru Space Engineering Forum 2026.

EDUCATION

BTech, Aerospace Engineering, Nilgiri Institute of Technology, Coimbatore (2016 to 2020). Captain of Team Sarus, the student rocketry team.

SKILLS

Propulsion	Thermochemistry (CEA), injector design, regenerative cooling, combustion stability, cold flow and hot fire testing, CFD (OpenFOAM)
GNC	Convex optimization, Kalman filtering, optimal control, 6-DOF simulation, attitude determination, TVC control, Monte Carlo analysis
Software	C++17, Python, MATLAB and Simulink, embedded Linux, FPGA basics, Git and CI
Test	Data acquisition, hardware in the loop, cryogenic systems, pressure systems, test conducting

CERTIFICATIONS

Hazardous Test Operations Conductor, Halyard Launch Systems (2023). Functional Safety for Flight Software, Avionics Assurance Academy (2024). Cryogenic Fluids Safety, Level 2, Southern Cryogenics Safety Board (2022). Pressure Systems Design and Safety, Deccan Engineering Institute (2021).